

ATMIYA DEVELOPER STUDENTS CLUB PRESENTS

CODE CARNIVAL 3.0

Idea Submission Template · The Heist Plan

TEAM NAME

Just Code

TEAM ID

NTQD

TEAM LEADER

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PROBLEM STATEMENT TITLE

PS-02 QueueLess Government Office

THEME / TRACK

E-Governance & Citizen Services

01 · THE TARGET

PROBLEM STATEMENT & UNDERSTANDING

PROBLEM STATEMENT

- Severe Physical Queues: Citizens wait 2 to 4 hours in crowded queues at Jan Seva Kendras & Taluka offices, often facing last-minute rejection at the counter due to minor document mistakes.
- Affected Citizens: Farmers applying for i-Khedut subsidies, students seeking scholarships, daily-wage laborers, and overworked desk officers facing counter bottlenecks.
- Massive Scale & Impact: Across Gujarat's 33 districts and 250+ talukas, over 40% of citizens make 2+ office trips for a single service, losing ₹500+ in daily wages per visit.

OUR UNDERSTANDING

- Root Cause: Zero pre-verification before office arrival. Citizens don't know exact statutory validity rules (e.g. 3-year income certificate limit, name mismatches, unverified bonafide).
- Gaps in Existing Portals: Traditional portals (Digital Gujarat / AnyRoR) provide static forms but lack automated AI authenticity checks, dynamic token queuing, and counter load balancing.
- Why Solve It Now: Digital Public Infrastructure (DPI) needs an intelligent pre-check gate so verified citizens complete their counter visit in under 5 minutes with zero rejection.

02 · THE PLAN

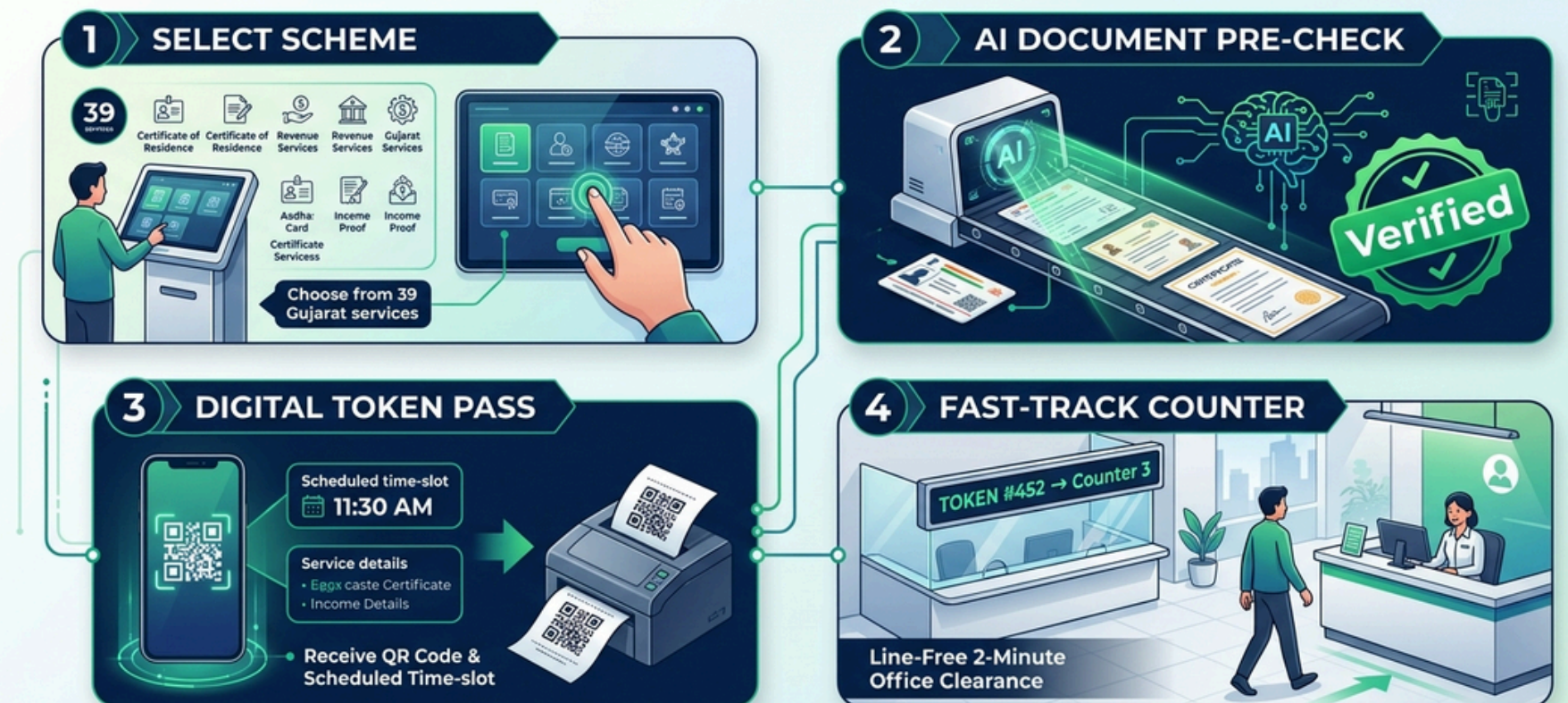
PROPOSED SOLUTION

OUR IDEA

- One-Line Solution: QueueLess Kacheri (NagrikSeva AI) — A Digital Public Infrastructure (DPI) web portal combining automated AI document pre-verification with smart time-slotted digital queue management.
- How It Addresses The Problem: Citizens pre-check and upload required paperwork from home; our multimodal AI validates authenticity and statutory rules, issuing a confirmed token pass with an exact arrival time.
- What Makes It Unique: In-browser Gemini AI Vision Officer — Automatically checks Gujarat-specific rules (3-year revenue validity, UIDAI seals, university bonafide certificates) with zero human backlog.
- Who Benefits: Citizens save 2–3 hours with line-free counter access; government counter operators receive 100% pre-verified applications, slashing counter time from 15 minutes to under 2 minutes.

QUEUELESS KACHERI E-GOVERNANCE PLATFORM

MODERN 4-STEP DIGITAL WORKFLOW



03 · THE ARSENAL

KEY FEATURES / FUNCTIONALITY

01 Multimodal AI Document Pre-Verification
Real-time Gemini Vision AI scans uploaded documents (Aadhaar, Bonafide, 7/12, Income Certificate). Enforces Gujarat statutory rules (3-year revenue validity, institutional stamps) and instantly rejects invalid receipts before office visit.

02 39 Gujarat Government Schemes Directory
Comprehensive bilingual catalog across Agriculture (i-Khedut), Health (PM-JAY), Education (MYSY/Namo Lakshmi), and Revenue with verified legal SLAs, zero/official fee transparency, and WhatsApp checklist sharing.

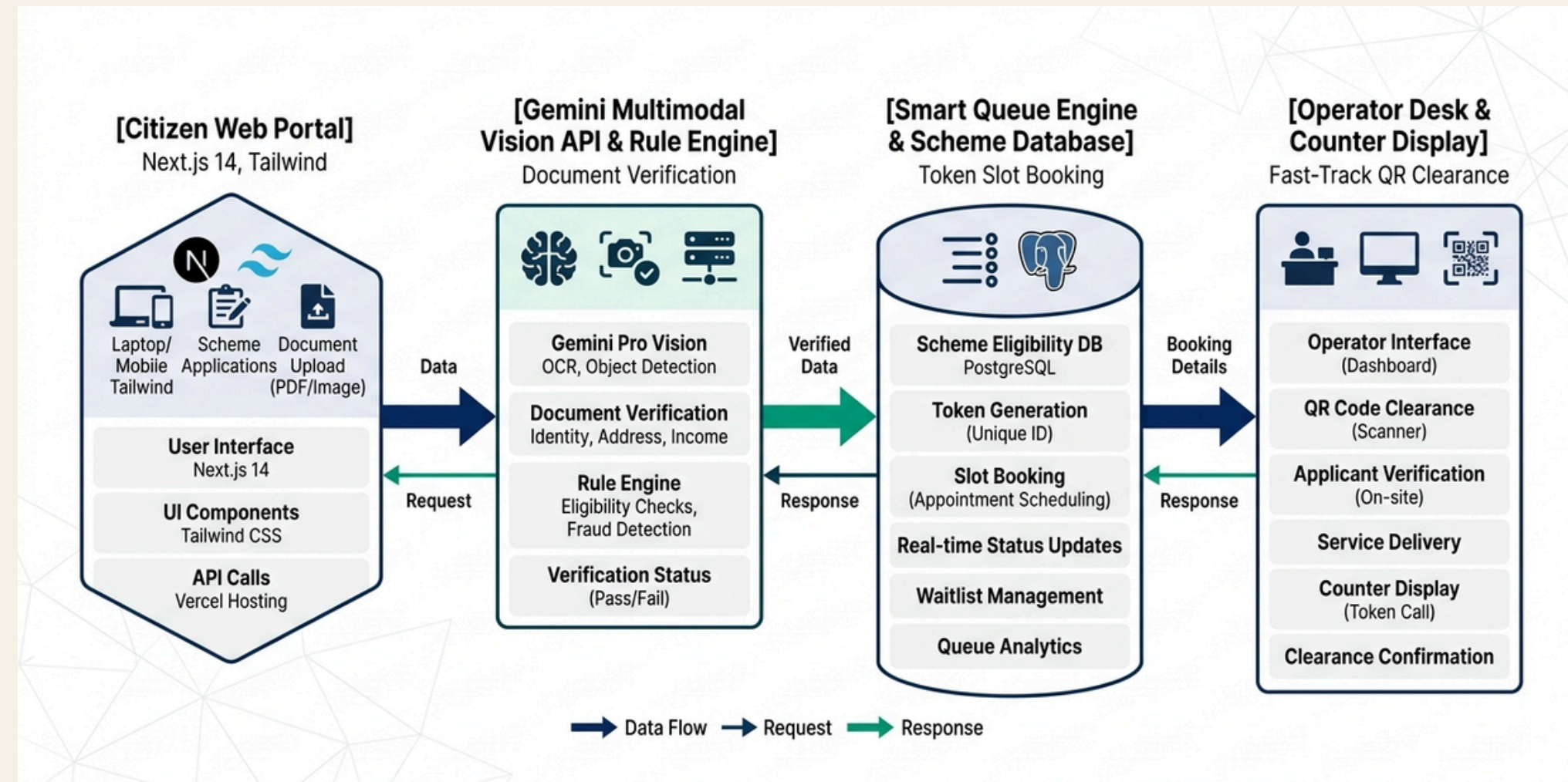
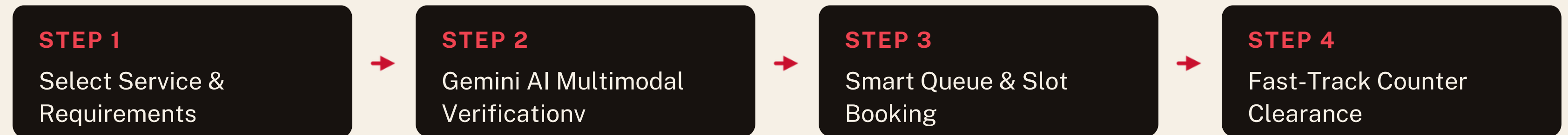
03 Scheduled Digital Token Pass (QR Code)

- Generates a cryptographically verified digital token pass with a precise arrival time-slot (e.g., 10:15 AM). Balances office crowd distribution and guarantees line-free fast-track counter entry.

04 Live Queue Radar & Mamlatdar Admin Desk
Citizens view live office congestion radars and counter wait times from home. Includes a dedicated Admin Desk for counter operators and Taluka officers to call tokens and manage citizen throughput.

04 · THE EXECUTION

HOW THE SOLUTION WILL WORK



05 · THE GEAR

TECHNOLOGY / TECH STACK

Layer	Technology	Why we chose it
Frontend	Next.js 14, React 18, Tailwind CSS, Lucide Icons	High-speed PWA performance, responsive civic UI, bilingual support
Backend	Next.js API Routes, Node.js (TypeScript)	Unified type-safe architecture, secure serverless execution
Database	Firebase Firestore & LocalStorage Cache	Real-time live queue synchronization, instant token state updates, offline citizen pass access
AI / APIs	Google Gemini 3.1 Flash-Lite Vision API & Canvas OCR	Sub-second multimodal OCR, strict fraud & document mismatch control
Hosting	Vercel Cloud Platform & Edge Network	Instant automated CI/CD deployment, 99.99% uptime, serverless scaling
Tools	GitHub, Postman, Figma, Google Antigravity IDE	[Reason in a few words]

06 · INSIDE THE MINT

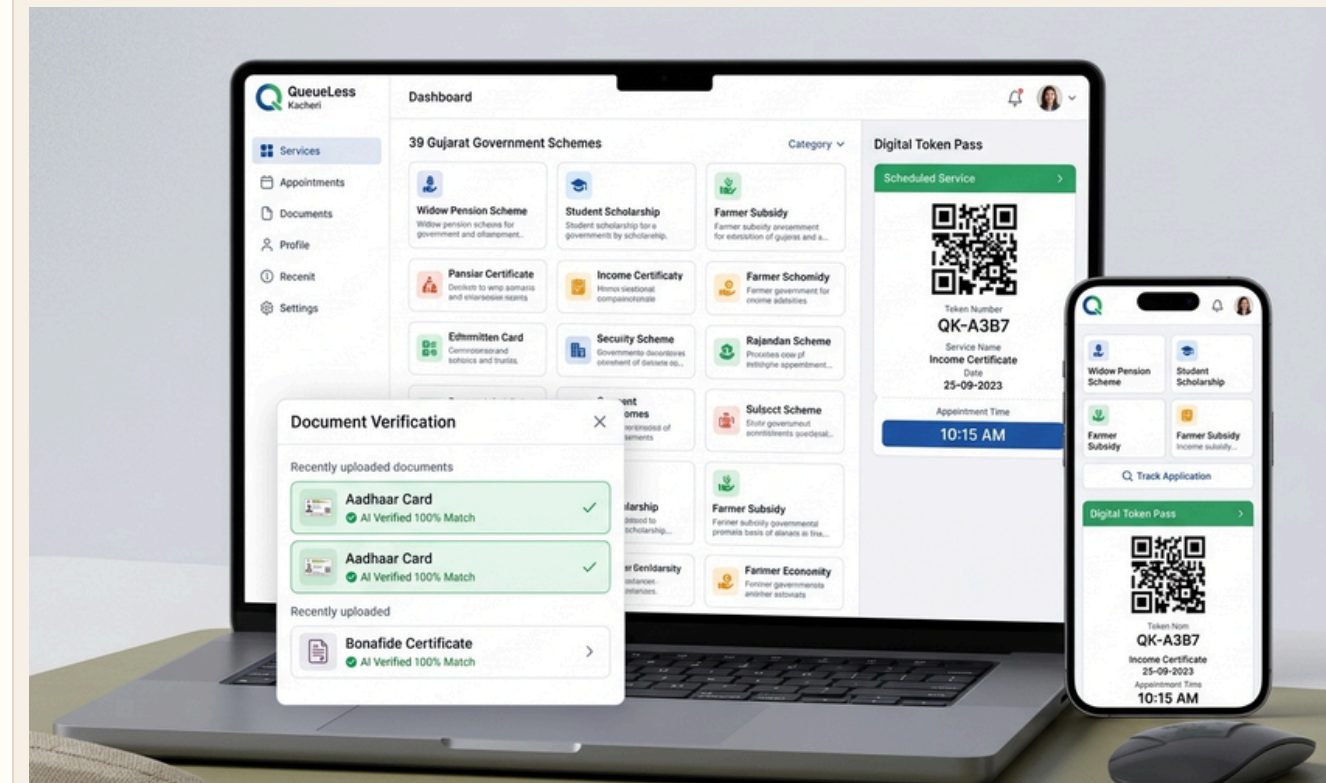
CURRENT PROGRESS / WORK DONE

COMPLETED

- 39 Gujarat Schemes Directory: Fully implemented catalog across Agriculture (i-Khedut), Health, Education & Housing with verified SLAs and checklists.
- Gemini Vision AI Pre-Verification: Live multimodal OCR validating Aadhaar, Bonafide certificates, 7/12, and 3-year revenue validity with zero fraud.
- Smart Queue & Slot Booking: 2FA civic authentication, time-slot allocation, and downloadable QR-coded Digital Token Pass.
- Operator Admin Desk: Working Mamlatdar and counter desk portal to call tokens and monitor office footfall.

IN PROGRESS

- Hybrid Govt Fee & Stamped e-Receipt: Integrating online UPI/Netbanking with counter cash options. Admin desk views real-time payment status alongside tokens and auto-issues an official digitally-stamped government receipt PDF upon verification.[What is partly working]
- DigiLocker Direct Cloud Sync: Building direct citizen cloud integration to auto-fetch verified certificates without manual paper scanning.
- District SLA Command Center: Designing live heatmap analytics for Taluka Mamlatdars and District Collectors to track queue clearing speeds.



07 · THE GETAWAY

FUTURE PLAN / NEXT STEPS

PHASE 1

DURING THE HACKATHON

- Feature Completion: Finalize hybrid fee payment gateway and automated digitally-stamped government receipt PDF generation.
- Testing & Edge Cases: Stress-testing Gemini AI Vision engine across 50+ document types including cropped photos, glare, and stamps.
- Final Demo Goal: Deliver a seamless live simulation: Citizen scheme booking -> AI document pre-check -> instant QR token call at Admin Desk.
- GPS Kacheri Radar & Office Locator: Integrated live geolocation finding the citizen's nearest Jan Seva Kendra / Taluka office with exact distance (km) and live crowd status.

PHASE 2

RIGHT AFTER

- DigiLocker & e-Pramaan SSO: Direct integration with national DigiLocker to auto-fetch verified Aadhaar, Ration, and 7/12 records without manual scanning.
- WhatsApp Bot & SMS Turn Alerts: Real-time automated messaging sending arrival reminders ("Your turn in 5 mins") and instant stamped PDF receipts.
- Entrance Kiosk & Thermal Token Printer: Self-service touchscreen kiosk mode for rural and walk-in citizens without smartphones.
- First Real Pilot in Rajkot: Live pilot launch at 1 Jan Seva Kendra / Mamlatdar office in Rajkot district to measure real wait-time drop.

PHASE 3

LONG-TERM VISION

- 18,000+ e-Gram Village Integration: Empowering Gram Panchayat VCE operators to pre-verify villager paperwork locally, ending taluka travel for rural citizens.
- Predictive Crowd AI & Staff Balancing: ML forecasting seasonal surges (e.g. scholarship rush in June, farm subsidy in Nov) to dynamically optimize counter staffing.
- Zero-Visit Paperless Delivery: Enabling 100% contactless digital issuance directly to citizen DigiLocker/WhatsApp for eligible pre-verified services.
- National Digital Public Infrastructure (DPI): Scaling the architecture across other Indian states under Digital India & Open GovStack frameworks.

08 · THE VAULT

GITHUB REPOSITORY LINK

 <https://github.com/hkPateL26/Queueless-government>

PUBLIC

- Repository is 100% public with open-source MIT license and clear directory structure.

README

- Complete documentation covering local setup, architecture diagrams, 39 schemes directory, and API specifications.

COMMITTS

Active development history with frequent commits, clean commit messages, and feature milestones.

DEMO (OPTIONAL)

https://drive.google.com/file/d/153-Ak-_nT-wYgv1F2lhbv4GfF17jr-9u/view?usp=sharing